Who Leaves Teaching, and Why?

Teacher Attrition in the United States, 2005–2025

José Elías Durán Roa[1]

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Abstract

American teachers leave the classroom at twice the rate the official statistics report. I document this by linking twenty years of Current Population Survey interviews taken twelve months apart, the first national data that follow every teacher into her next occupation or out of the labor force. The official rates survive intact inside my data once their universe and definitions are imposed; the other half of the flow is the margin no official instrument can see, teachers who move directly into other jobs. These moves account for the entire rise in attrition of the past two decades, a third of them never leave education, and they mark who departs, since part-time hours, private schools and lower pay predict exit while demographics barely matter. Judged against seven professions of similar composition under the identical design, teacher turnover is unexceptional, and teachers leave teaching more slowly than nurses leave nursing. What sets teaching apart is motherhood. A new baby raises a teacher's probability of leaving the profession by six percentage points, the same event moves no comparable profession, and fathers are unaffected everywhere. Teaching does not have an exodus problem. It loses its new mothers.

[1]PhD Candidate in Economics, University of Alicante, and Research Intern at the OECD Directorate for Education and Skills. E-mail joseelias.duranroa@ocde.org and joduran@ua.es. The views expressed here are mine and do not represent those of the OECD. All remaining errors are my own.

1. Introduction

Few questions in the economics of education carry more policy weight than who leaves teaching, and why. Teachers are the main school input in the production of achievement (Rivkin, Hanushek and Kain, 2005; Chetty, Friedman and Rockoff, 2014), replacing a leaver is expensive and disruptive for the students left behind (Ingersoll, 2001; Ronfeldt, Loeb and Wyckoff, 2013), and the fear of a post-pandemic exodus has moved retention to the center of the education policy debate. Yet the national statistics can say how many teachers leave, and little about where they go or why, because no national source follows a teacher into whatever she does next.

What is known comes from three kinds of sources, and they agree on the level while sharing that blind spot. The Teacher Follow-up Survey puts the official leaver rate at 8 percent of public school teachers in 2021–22, but it runs only every few years and observes leavers in a single retrospective questionnaire (NCES, 2024). State administrative records support the sharpest causal work (Hanushek, Kain and Rivkin, 2004; Biasi, 2021; Bacher-Hicks, Chi and Orellana, 2023; Goldhaber and Theobald, 2023; Camp, Zamarro and McGee, 2025), but each covers one state and loses sight of a teacher the moment she leaves the state payroll. The existing national estimates from the CPS rely on the March supplement question about the job held one year earlier and land in the same range (Harris and Adams, 2007; Aldeman and Yi, 2025). All three see a teacher leave; none can see where she goes.

I build the missing source from the same survey. The CPS re-interviews every address twelve months apart, and by linking those interviews I follow every teacher it observes between 2005 and 2025, public or private, full or part time, into her next occupation or out of the labor force, 174,872 linked observations of 54,944 distinct teachers. The design answers a plain question, the share of the people teaching school this year who will no longer be teaching school next year, and what they do instead. Because one in six apparent leavers is back in a classroom within three months, my main measure counts as a leaver only a teacher who is out of teaching at the twelve-month mark and in every re-interview after it.

Three findings organize the paper.

The first is about how much. The classroom loses roughly twice as many teachers as the leaver statistics count, 15.1 percent per year, up from about 13 percent in the mid 2000s to 17 in recent years. The official numbers survive intact inside my data once their universe and definitions are imposed; the gap is the margin their instruments compress, teachers who move directly into other jobs. That same margin accounts for the entire twenty-year rise, so the rise is a reallocation rather than a withdrawal, and 30 percent of leavers never leave education.

The second is about who departs, and through which door. Exit concentrates among the loosely attached, since part-time hours, private schools and lower pay predict leaving while demographics barely matter, and the doors are ordered. Better options pull teachers into other occupations, which is where all the growth is, while family events push mothers, and only mothers, out of the labor force.

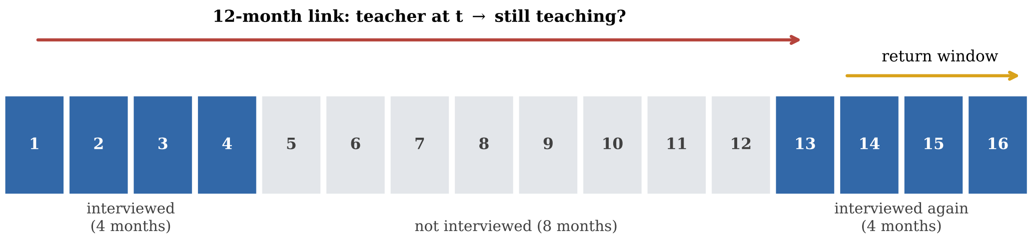
The third is about what is peculiar to teaching, and it is not the amount, since under the identical design teachers leave teaching more slowly than nurses leave nursing or pharmacists leave pharmacy. It is the motherhood door. A new baby raises a teacher's probability of leaving the profession by 6 percentage points, the national counterpart of the cohort evidence in Stinebrickner (2002), while the same event moves no other profession I can follow and fathers are unaffected everywhere.

None of the estimates is causal; they are the descriptive national infrastructure that a causal literature built on single states currently lacks (Nguyen, Pham, Crouch and Springer, 2020). Section 2 describes the data, Section 3 who teaches, Section 4 how many leave, Section 5 where they go, Section 6 who leaves and why, Section 7 places teaching among the professions, and Section 8 concludes with policy. Appendix A reconciles my series with every published benchmark; Appendix B holds the full estimation output.

2. Data and definitions

I use the basic monthly microdata of the Current Population Survey, the household survey behind the official employment statistics of the United States, assembled for every month between January 2005 and December 2025.[2] Each household is interviewed during four consecutive months, rests for eight, and is interviewed again during four more (Figure 1), so an adult interviewed in year t is re-interviewed in the same calendar month of $t+1$. I link the two interviews through household and person identifiers, validating every match on sex, race and a plausible age progression (Madrian and Lefgren, 2000); linking the public files this way is how the Bureau of Labor Statistics builds its official labor force flows and how IPUMS builds its longitudinal keys (Rivera Drew, Flood and Warren, 2014). Teachers are identified by the occupation of their main job, classroom teaching from preschool through secondary school including special education,[3] with at least a bachelor's degree, the standard restriction in this literature.

Figure 1: The CPS follows each person for at most 16 months.



Notes: Interview calendar of one rotation group; numbers are months since the first interview. The 12-month link compares an interview from the first block with the interview held exactly one year later, and the return window uses the remaining interviews of the second block.

The object I measure is deliberately plain. Of everyone teaching school in a given month, public or private, full or part time, I ask who is no longer teaching school twelve months later, and what she is doing instead. Temporary exits such as parental leave would inflate that count, and indeed 15.7 percent of 12-month leavers are teaching again within three months, so my main outcome is the non-returning leaver, a teacher out of teaching at $t+12$ and in every later re-interview, computed on the 129,897 baseline observations, from 52,238 distinct teachers, whose rotation allows at least one re-interview. A person contributes up to four linked baselines and standard errors are clustered by household throughout, which nests the person. Appendix A traces the population funnel, shows that linkage does not select against teachers, reconciles my rates with every published benchmark, and states the caveats that make my levels upper bounds on permanent exit.

[2] The files come from the U.S. Census Bureau for 2024–2025 and from the NBER mirror for 2005–2023. October 2025 was never released because the federal government shutdown suspended that month's data collection.

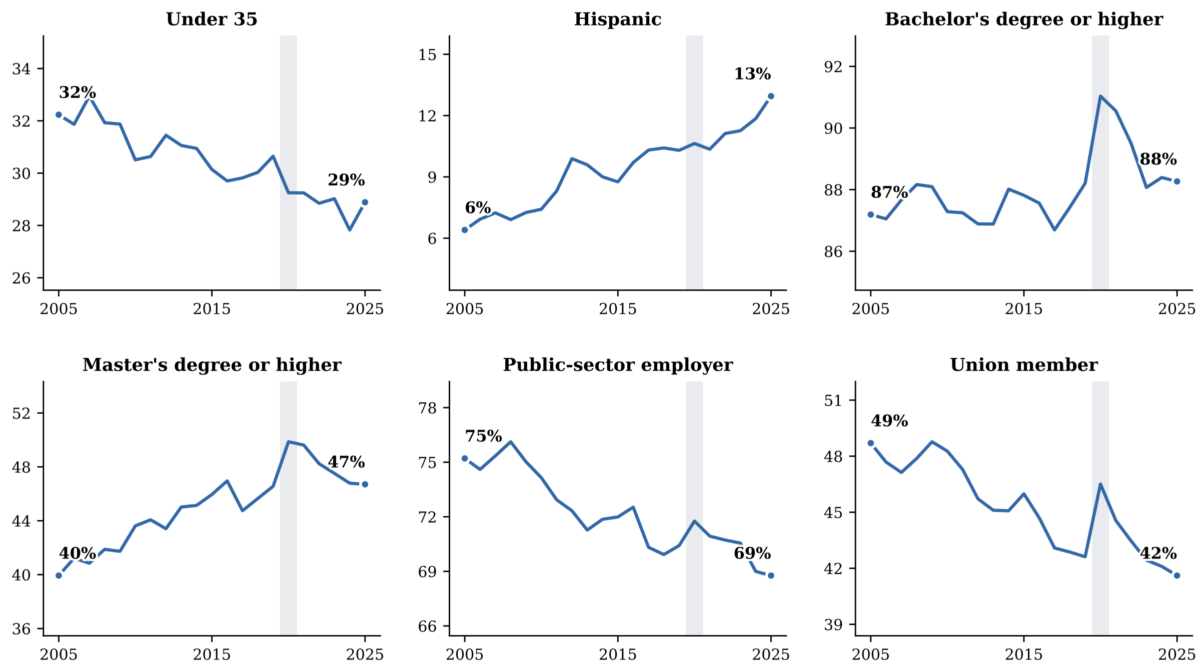
[3] Census occupation codes 2300 to 2330, identical in the 2002, 2010 and 2018 classifications used over the period.

3. Who teaches

Before asking who leaves, it helps to know who teaches. Table 1 draws that portrait on every person I observe in a school-teaching occupation over the two decades, 116,649 individuals who represent about 5.2 million teachers in an average month, split by sex and set against the rest of the college-educated workforce; the portrait needs no degree restriction and no successful link, which is why its universe is wider than the 54,944 teachers of the attrition panel. The typical teacher is a married woman in her early forties with children at home. Teaching is overwhelmingly female, at 78 against 47 percent, disproportionately white, and markedly less open to Asian and foreign-born graduates. Credentials cluster in the middle, with 45 percent holding a master's degree, ten points above other graduates, while doctorates remain rare. Almost three quarters work for the public sector against 15 percent elsewhere, 45 percent are union members against 9, and the price of this stability is pay, since the median teacher earns $1,154 a week, roughly a quarter less than comparable college-educated workers.

The profile has been moving, and Figure 2 traces that movement year by year. The workforce is aging, with the share under 35 down from 32 to 29 percent, becoming markedly more Hispanic, from 6 to 13 percent, and steadily more credentialed, with master's degrees rising from 40 to 47 percent, while two anchors of the traditional teaching career erode at the same time, public-sector employment falling from 75 to 69 percent and union membership from 49 to 42. The gender mix, in contrast, has not moved at all. One of the eroding anchors matters directly for what follows, because the public sector is precisely the trait that Section 6 will show holds teachers in place, so the workforce has been drifting toward the configuration that leaves.

Figure 2: The changing profile of the teaching workforce.



Notes: Weighted annual means over all employed persons observed in a school-teaching occupation, with no degree restriction. Union membership is only asked in the outgoing rotations. The shaded band marks 2020. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Table 1: Portrait of the American teaching workforce, 2005–2025.

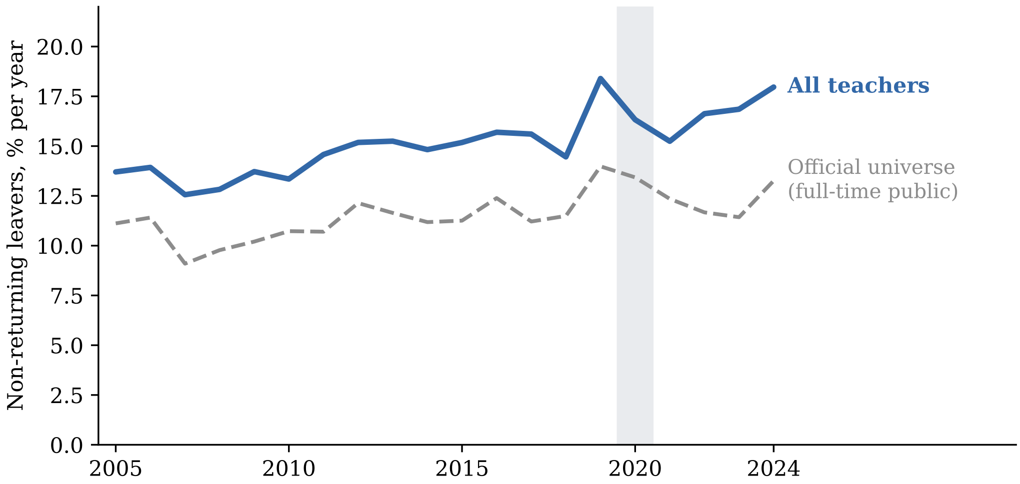
	School teachers			Other college-educated workers	Difference
	Women	Men	All		
Panel A. Demographics					
Age, years	42.6	43.1	42.7	43.8	-1.11***
Female			78.2%	47.3%	30.9***
White	84.8%	85.3%	84.9%	78.5%	6.4***
Black	10.3%	9.8%	10.2%	9.0%	1.2***
Asian	2.8%	2.6%	2.8%	10.3%	-7.5***
Hispanic	9.4%	9.3%	9.4%	8.3%	1.0***
Non-citizen	2.2%	2.3%	2.3%	7.2%	-4.9***
Panel B. Family					
Married	65.8%	69.1%	66.6%	63.8%	2.7***
Number of own children (<18) at home	0.8	0.8	0.8	0.6	0.15***
Child under 6 at home	16.7%	19.9%	17.4%	15.6%	1.8***
Panel C. Education					
Bachelor's degree or higher	86.2%	94.4%	88.0%	100.0%	-12.0***
Master's degree or higher	43.5%	50.8%	45.1%	34.5%	10.6***
Professional degree or doctorate	2.1%	3.9%	2.5%	10.5%	-8.0***
Panel D. The job					
Public-sector employer	70.5%	78.0%	72.1%	15.4%	56.8***
Union member[a]	43.9%	50.2%	45.3%	8.7%	36.6***
Usual weekly hours	39.5	41.4	39.9	40.4	-0.49***
Part-time (<35 h/week)	11.7%	6.8%	10.6%	12.3%	-1.7***
Holds more than one job	6.4%	11.3%	7.4%	6.1%	1.3***
Panel E. Pay and income					
Weekly earnings, median[b]	$1,115	$1,308	$1,154	$1,500	−$346***
Family income $75k+[b]	81.1%	84.7%	81.9%	85.5%	-3.6***
Persons	92,044	24,812	116,649	963,782	
Population represented, millions[c]	4.1	1.1	5.2	49.6	

Notes: Weighted means over every employed person observed in a school-teaching occupation, with no degree restriction, pooling all 251 survey months; the comparison group is employed persons outside teaching holding at least a bachelor's degree. Sample sizes are unique persons, and a small number of persons with inconsistent sex codes across interviews appear in both sex columns. The last column reports the difference between all teachers and the comparison group, in percentage points for shares, and the significance of a two-sample test with standard errors clustered by household ($^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$). [a] Union membership and weekly earnings are only asked in the outgoing rotations, months in sample 4 and 8. [b] Measured over 2021–2025 only, to avoid mixing twenty years of price levels. [c] Population represented by the group in an average month. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

4. How many leave

More teachers leave than the official statistics count, under every definition, and the loss is rising under every definition. Counting everyone, attrition net of returns climbed from about 13 percent in 2005–2010 to about 17 percent in 2022–2024, with a new high of 18.0 percent in 2024 (Figure 3). Restricted to the official universe of full-time public school teachers, the series runs about three and a half points lower year after year and climbs in parallel, so nothing in the trend depends on private schools or part-timers; what they add is level, because they are exactly the teachers who leave most, and every model estimate of Section 6 survives dropping them (Appendix Table A7, column 5).

Figure 3: Attrition net of returns, all teachers and the official universe.



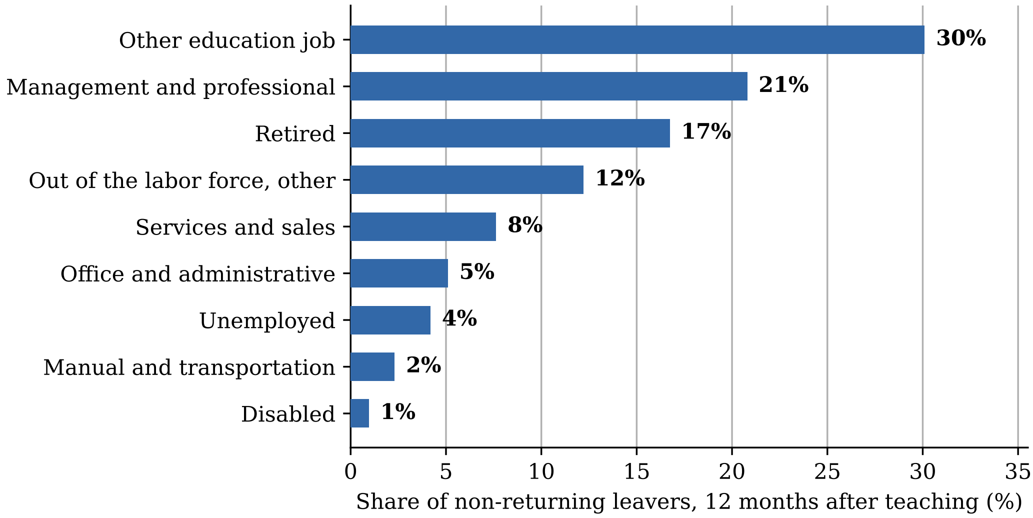
Notes: Weighted share of teachers not teaching twelve months later and in no later re-interview, by baseline year. The official universe keeps full-time public school teachers above the preschool level, the population covered by the Teacher Follow-up Survey. The gap between the two series averages 3.6 percentage points with a standard deviation of 0.8. The shaded band marks 2020. Source: author's calculations from linked CPS pairs, 2005–2025.

That level is roughly twice the 8 percent of the official statistics, and the distance is definitions, not disagreement. Imposing the official universe lowers my rate from 15.1 to 11.5 percent, and treating every leaver found in another education occupation a year later, tutoring, teaching assistance, school administration, as if she had never left the classroom, the most conservative possible reading of the occupation codes, lowers it to 8.0, the official number itself. The official 8 is therefore nested in my data between two transparent choices, and Appendix A walks every step with its sample size. My headline stays at 15.1 percent because the paper asks the broader question, how many of the people teaching school this year will not be teaching school next year.

5. Where they go

Leavers do not go far, and they rarely go out of work. The single largest destination, at 30 percent of non-returning leavers, is another education job (Figure 4), whether postsecondary teaching, tutoring and instruction, teaching assistance, librarianship or school administration, so a large share of exits from classroom teaching never leave education at all. Retirement accounts for a further 17 percent, at a mean age of 62, and only 4 percent are unemployed a year later. The gender contrast is sharp exactly where the model of Section 6 will point, since 14 percent of women who leave are out of the labor force for reasons other than retirement or disability, at a mean age of 39 consistent with family care, against 5 percent of men, while men move more often into management and professional jobs, 26 against 19 percent.

Figure 4: Situation of non-returning leavers twelve months after teaching.

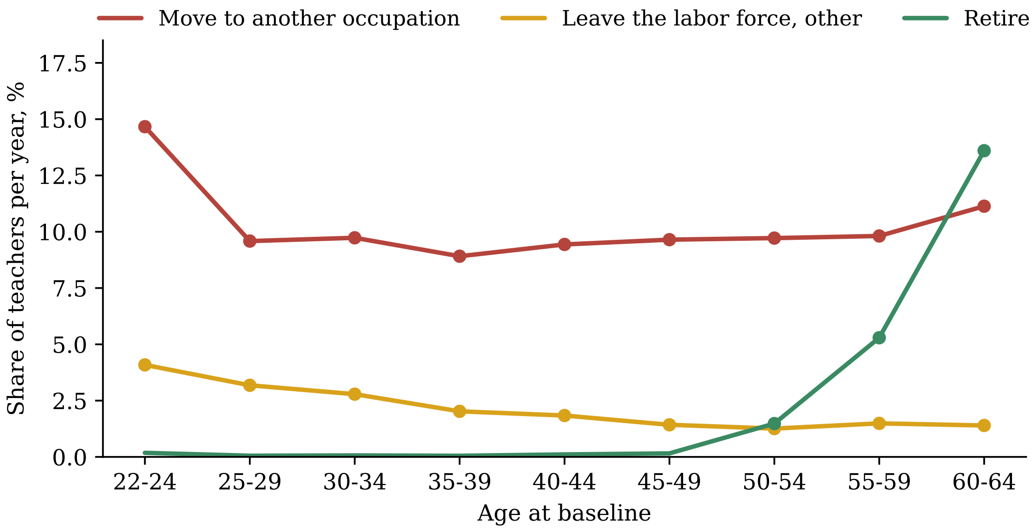


Notes: Education jobs other than school teaching include postsecondary teaching, tutoring and instruction, teaching assistance, librarians and education administration, and the remaining employed leavers are grouped into broad occupation classes. Retirement, disability, unemployment and other non-participation come from the labor force status recorded at the second interview. Source: author's calculations from linked CPS pairs, 2005–2025.

The rise of Section 4 has a single engine. It is almost entirely a rise in occupation switching, from about 8.5 percent of teachers per year in the mid 2000s to 12 percent in 2022–2024, while exits into non-participation have stayed flat at 4 to 5 percent and unemployment spikes only in recessions (Appendix Figure A1). Teaching is increasingly losing people to other jobs, not to inactivity. Mobility inside the profession is substantial as well, since among teachers still teaching a year later 6.7 percent have moved from a public to a private school and 5.8 percent in the opposite direction.

Age reorganizes the same routes completely (Figure 5). Moving to another occupation is not a young teacher's phenomenon, it is the dominant route at every age below sixty, holding remarkably flat at 9 to 10 percent of teachers per year from the late twenties through the fifties. What changes over the lifecycle is everything else. Family and other labor-force exits fade with age, from 4.1 percent of teachers at ages 22 to 24 to 1.4 in the late forties, while retirement is invisible before fifty and then takes over, reaching 13.6 percent per year at ages 60 to 64. A district worried about mid-career losses is worried, whether it knows it or not, about competition from other employers.

Figure 5: Exit routes over the lifecycle.



Notes: Weighted share of teachers leaving through each route within twelve months, by age at baseline, pooling all 2005–2025 waves under the non-returning definition. Routes come from the labor force status recorded twelve months later; the labor-force route excludes retirement, shown separately, and includes disability. The unemployment route never exceeds 1.2 percent of teachers at any age and is omitted. Markers are drawn only where a route reaches one percent of teachers per year. Source: author's calculations from linked CPS pairs, 2005–2025.

6. Who leaves, and why

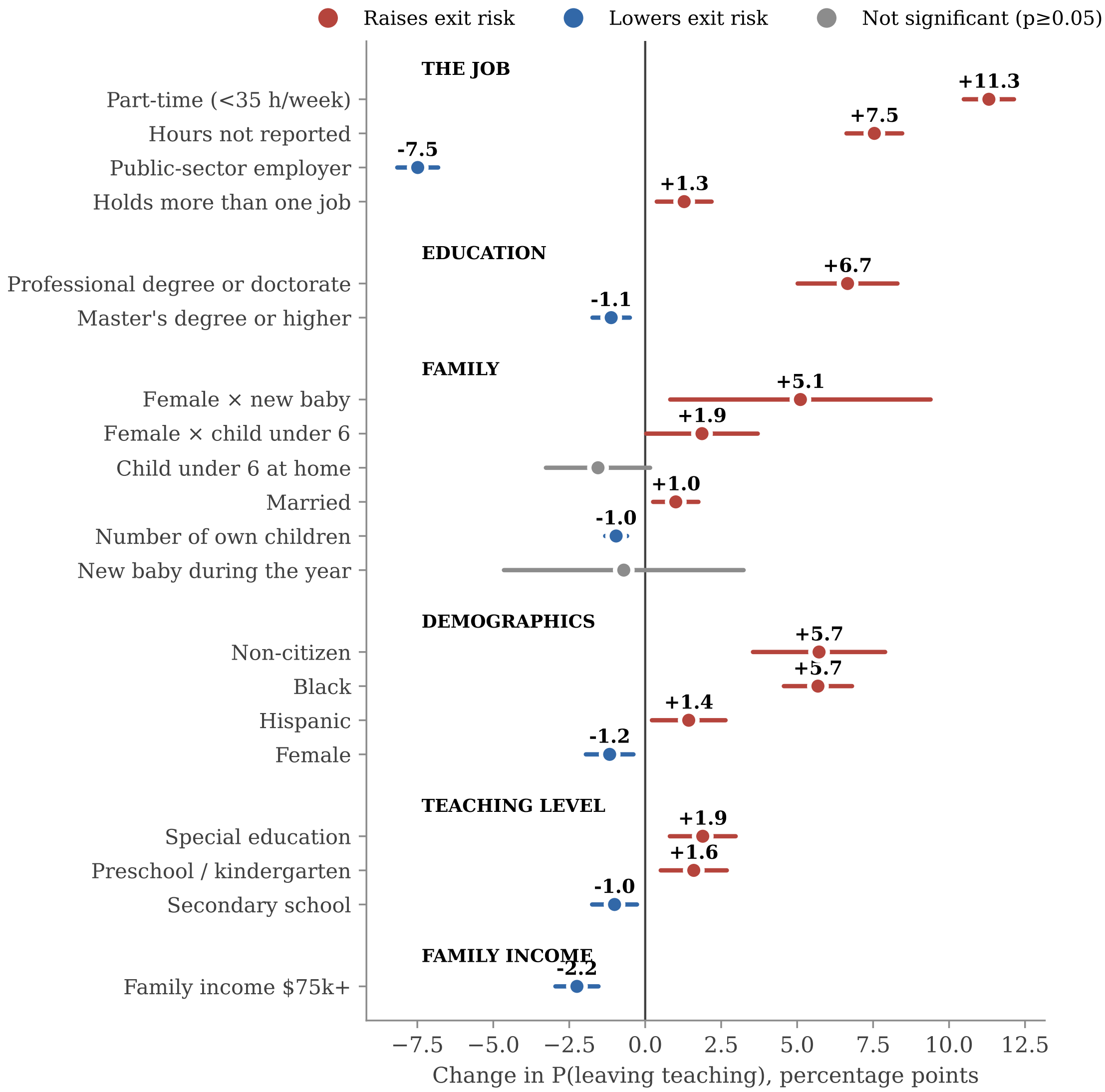
Leaving concentrates among the loosely attached, and the raw contrasts say so before any model. Compared with stayers (Appendix Table A5), leavers are three years older, working part-time is eleven points more common among them and private-sector employment fifteen points more common, and they have fewer young children at home rather than more, while the gender split is economically irrelevant, 76.8 percent of stayers are women against 77.0 of leavers. Expressed as exit rates, part-time teachers leave at more than twice the rate of full-time teachers, 32 against 13 percent, and preschool and kindergarten teachers face 21 percent against 15 for elementary and 13 for secondary school.

To weigh every characteristic at once, I estimate a probit for the probability that teacher i, observed in year t, leaves without an observed return,

$$\Pr(\text{leave}_i = 1 \mid x_i) \;=\; \Phi\big(x_i'\beta + \gamma_t\big)\,, \tag{1}$$

where x_i collects demographics, family structure, credentials, hours, sector and region at baseline and γ_t are base-year fixed effects. I report average marginal effects in percentage points, with standard errors clustered by household; these are conditional associations, not causal effects. Figure 6 plots the estimates and Appendix Table A8 reports the full model.

Figure 6: Average marginal effects on the probability of leaving teaching, with 95% confidence intervals.



Notes: Estimates from the probit of Table A8. The age profile is shown in Appendix Figure A2. Region indicators and the hours-not-reported indicator are included in the model but not displayed.

The job margin dominates. A part-time teacher is 11.3 percentage points more likely to leave than an otherwise identical full-time colleague (Appendix Table A8, column 2, significant at 1 percent), and a private-school teacher is 7.5 points more likely to leave than a public-school one; against a baseline exit rate of 15 percent, the two largest effects in the model amount to roughly three quarters and one half of the average risk. Pay retains, modestly. On the outgoing-rotation subsample of 44,823 teachers with observed earnings, ten percent higher weekly pay lowers the probability of leaving by 0.3 percentage points, about two percent of the baseline rate.

The family results separate stocks from events. Each own child at home predicts about one point less exit (significant at 1 percent), while the event of a birth works in the opposite direction and only for women. A new baby leaves men's exit probability unchanged, a precisely estimated zero with a p-value of 0.72, but raises women's by 5.1 points, significant at 5 percent and one third of the baseline risk, and 58 percent of the new mothers who leave exit the labor force entirely against 30 percent of the average leaver. Black teachers leave at 5.7 points above comparable white teachers and non-citizens at 5.7 points above citizens, both significant at 1 percent, gaps that compound into the diversity of the future workforce. Credentials cut both ways, since a master's retains modestly while a professional degree or doctorate predicts 6.7 points more exit, consistent with better outside options, and the predicted probability of leaving follows a U in age, with its minimum of 9 percent around age 39 (Appendix Figure A2).

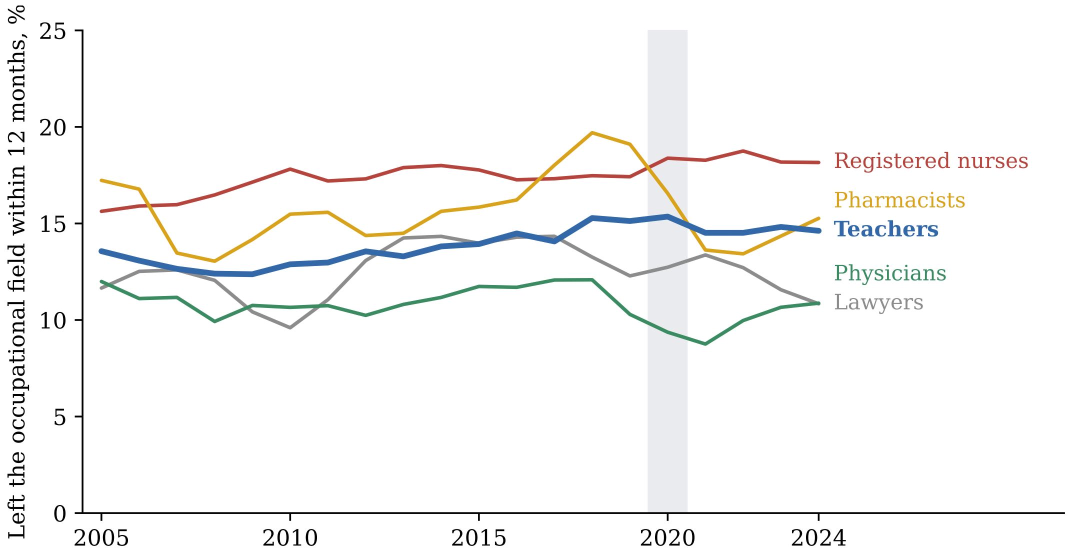
None of this depends on the estimator, the period or the part-timers. A logit, a linear probability model, dropping the pandemic years 2019–2021, or estimating on full-time teachers only leaves every effect essentially unchanged (Appendix Table A7), and the same model on subsamples shows the job effects are universal, for women and men, public and private, younger and older (Appendix Figure A3). The routes tell more than the total. Estimating the model separately for each exit route, the national counterpart of Stinebrickner (2002) and Dolton and van der Klaauw (1999), splits the predictors into two clean families (Appendix Table A6). The opportunity variables work almost entirely through job switching, since the public-sector advantage is 6.4 of its 7.5 points on the occupation margin and a professional degree or doctorate raises switching by 6.3 points while leaving labor-force exit untouched. The family variables work almost entirely through non-participation, since a new baby moves mothers out of the labor force by 6.6 points while reducing their job switching by 2.3. Pay retains on both margins.

7. Teaching among the professions

One in seven means little until it is judged against how fast similar people leave similar careers, so I build the identical linked panels for seven other degree-holding professions, registered nurses, pharmacists, physical therapists, physicians, lawyers, social workers and accountants. On the amount, teaching is unexceptional. Figure 7 plots exits from the whole occupational field for the five professions with distinctive, cleanly coded titles, counting moves inside a field, a teacher promoted into school administration or a staff nurse becoming a nurse practitioner, as staying, and teachers leave teaching more slowly than nurses leave nursing or pharmacists leave pharmacy in every single year, faster only than physicians and lawyers, whose training costs most. No profession in the figure shows a post-pandemic break. Whatever problem teaching has, it is not an exodus, the same conclusion Harris and Adams (2007) and Aldeman and Yi (2025) reach from the retrospective March question; Appendix Table A4 reports my replication of their design next to the linked rates for all seven professions.

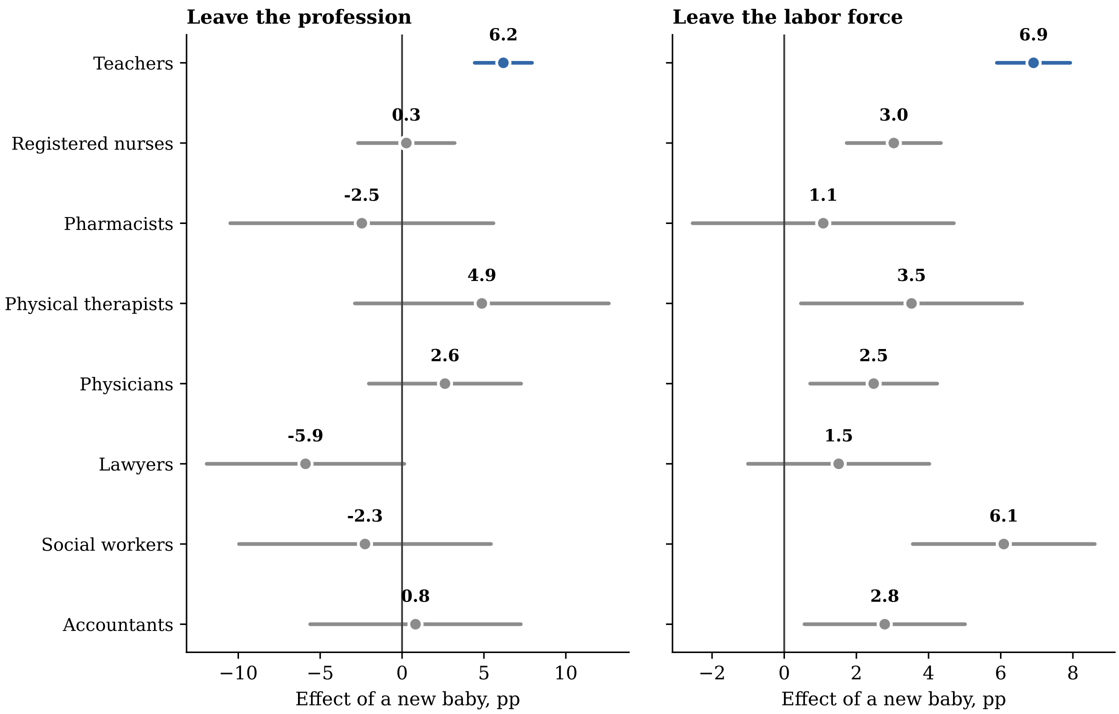
The exception appears on the family margin. Section 6 showed that a birth pushes mothers, and only mothers, out of teaching, but that fact alone cannot say whether the push is about teaching or simply about motherhood, since new mothers step back from work in every occupation. To separate the two, Figure 8 runs the identical birth event through every profession, the profession-level counterpart of the child penalty literature (Kleven, Landais and Sogaard, 2019; Gulek, 2025), and the push turns out to be about teaching. A new baby raises a teacher's probability of leaving her profession by 6.2 percentage points (standard error 0.9), the only significantly positive effect among the eight professions. The mechanism is in the second panel. New mothers pause work everywhere, in nursing by 3.0 points, but the other professions absorb the pause without losing the woman from the field, while teaching converts nearly its entire labor-force exit of 6.9 points into a loss for the profession. Nursing, where shift work makes hours genuinely divisible, loses none of its new mothers, the same part-time mechanism that Goldin and Katz (2016) document for pharmacy, so the profession most often described as family friendly is the one a birth is most likely to end a career in. Fathers show no effect anywhere, in teaching or outside it.

Figure 7: Leaving the occupational field within twelve months, five professions under the same linked design.



Notes: Weighted shares of employed college graduates no longer employed in any occupation of their field twelve months later, education occupations for teachers, health-practitioner occupations for nurses, pharmacists and physicians, and legal occupations for lawyers, linked and validated exactly as the teacher panel and smoothed with a centered three-year moving average. Linked observations are 174,872 teachers, 75,160 nurses, 45,294 lawyers, 33,367 physicians and 10,449 pharmacists. The shaded band marks 2020. Appendix Table A4 reports all seven comparison professions under both the linked and the retrospective instruments. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure 8: A new baby and the probability that a woman leaves her profession.



Notes: Average marginal effects of a new baby, an own child under three appearing between the two interviews, from probits estimated on women of each profession with age, age squared, marital status, an advanced-degree indicator and base-year fixed effects, with 95% confidence intervals and standard errors clustered by household. Leaving the profession is leaving the whole occupational field under the conventional 12-month definition, so a move inside education or inside health practice is not an exit; the teacher effect of 6.2 points sits next to the 5.1 of the non-returning interaction in Appendix Table A8. Births observed among women range from 160 for physical therapists to 3,212 for teachers. Source: author's calculations from linked CPS pairs, 2005–2025.

8. What it means for policy

The results locate attrition in weak attachment to the job, in part-time hours, private-sector positions and both career extremes, rather than in broad demographics, with a rising trend made almost entirely of moves to other occupations; the teacher at highest predicted risk is a late-career woman working part-time outside the public sector, not the young idealist of the burnout narrative. Four levers follow directly. Converting involuntary part-time positions to full-time attacks the single largest risk factor, and the wage result implies that pay raises retain, though modestly. The private sector is where teaching careers break. The Black, Hispanic and non-citizen retention gaps deserve monitoring for the diversity of the future workforce. And with one in six leavers back within three months and a third of the rest still working in education, cheap re-entry pathways such as fast-track rehiring and return bonuses can recover part of the outflow at low cost, because most leavers never really left the field.

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Appendix A. Sample, representativeness and reconciliation

Table A1 traces how the CPS universe narrows down to my analytic sample. In an average month the survey interviews close to 95,000 adults who represent 241 million people, of whom about 1,850 are employed school teachers with at least a bachelor's degree, representing the 4.6 million teachers of the country. Only the half of the sample that is in its first rotation block can be linked one year ahead, and the linkage succeeds for 78 percent of teachers. Reassuringly, it succeeds less often for everyone else, 74 percent among other college-educated workers and 71 percent among all employed, a ranking that holds in every single year, so sample losses do not select against teachers. Losses arise because the CPS samples addresses rather than people, which means movers drop out, and since moving is correlated with changing jobs my attrition rates are conservative on this margin. Linked persons keep their population weights, so all weighted rates remain representative, and Table A2 confirms that the linked panel matches the full teacher workforce on every observable.

Table A1: From the CPS universe to the analytic sample.

	Persons	Population represented
Adults interviewed in an average month	94,630	240.6M
employed	57,572	148.1M
school teachers (occ. 2300–2330)	2,094	5.2M
with bachelor's degree or higher	1,849	4.6M
Linked 12 months later (54,944 teachers, 20 waves)	174,872	the same 4.6M
with re-interviews after $t+12$: **main sample**	129,897	the same 4.6M

Notes: Monthly figures are averages over the 251 available months, with weighted population totals beside the sample counts. Teachers are identified by Census occupation codes 2300 to 2330 in the main job and a bachelor's degree or higher. The linked rows require a validated match to the interview twelve months ahead, and the last row keeps the baselines whose rotation allows at least one re-interview after the link. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Table A2: Composition of the linked panel vs all teacher interviews.

	All teacher interviews	Linked panel
Age, years	43.0	44.0
Female	76.6%	76.8%
White	86.2%	87.6%
Master's degree or higher	51.2%	52.9%
Public-sector employer	76.8%	78.3%
Part-time (<35 h/week)	8.7%	9.0%
Unique persons	98,972	54,944

Notes: Weighted means. The first column pools every monthly interview of an employed, degree-holding school teacher between 2005 and 2025; the second column is the baseline of the linked 12-month panel used throughout the paper. Source: author's calculations from CPS basic monthly microdata, 2005–2025.

Two measurement caveats apply throughout. First, the observation window is short by design. A person leaves the survey for good at most sixteen months after entering it, so a teacher who returns to the classroom more than a few months after the twelve-month link is still counted as a leaver, and all my attrition rates are upper bounds on permanent exit. Second, occupation in the returning rotation is re-coded independently, which mechanically inflates measured switching (Mellow and Sider, 1983; Kambourov and Manovskii, 2013), so levels should again be read as upper bounds while comparisons across groups and over time, the object of this paper, are far less affected.

My rates contain every published benchmark. Table A3 walks the distance step by step, and every step is an institutional difference between the sources rather than an adjustment. The Teacher Follow-up Survey samples K–12 classroom teachers from school rosters, explicitly excluding short-term substitutes, student teachers and teacher aides, and reports that 8.0 percent of public school teachers left the classroom between 2020–21 and 2021–22 (NCES, 2024). My universe is instead anyone whose main job is school teaching, substitutes included. Purging observed returns lowers my conventional 17.6 percent to 15.1, restricting to public K–12 teachers lowers it to 12.7, dropping part-timers, the closest observable proxy for the substitutes that the rosters exclude, lowers it to 11.5, and counting as leavers only teachers found in no education occupation at $t+12$, the margin where independent occupation coding is most fragile, yields 8.0. This last step deliberately overshoots, because the follow-up survey counts moves into non-teaching education jobs as leaving while my strictest row does not, so the fully aligned figure lies between 8.0 and 11.5 and the survey benchmark sits inside that bracket. The private side tells the same story from the other direction, since the follow-up survey itself reports 12.0 percent attrition in private schools, and its leavers are not gone from education either, as 39 percent of them keep working for a school or district in a non-teaching role, the same phenomenon as the 30 percent of education destinations in Figure 4.

Table A3: From this paper's definitions to the published benchmarks.

	Annual rate	Observations
Conventional 12-month leaver, all teachers	17.6%	174,872
Non-returning leaver (main definition)	15.1%	129,897
public K–12 teachers only	12.7%	98,122
and full-time only	11.5%	91,035
and in no education occupation at $t+12$	8.0%	91,035
Benchmarks and retrospective measures		
Teacher Follow-up Survey, public school leavers	8.0%	∼10,300[†]
Teacher Follow-up Survey, private school leavers	12.0%	∼10,300[†]
My retrospective replication, 2022–2024 March CPS	5.8%	6,945
without the degree restriction	6.7%	7,909
Harris and Adams (2007); Aldeman and Yi (2025)	7.7; 7.6%	not reported

Notes: Each row applies one restriction on top of the previous one to the linked panel under the non-returning definition, on the subsample whose rotation allows re-interviews, restricting it step by step to the Teacher Follow-up Survey universe; full-time status is the observable proxy for the roster-based exclusion of short-term substitutes, and the final row counts as leavers only teachers found in no education occupation twelve months after baseline. The benchmark rows quote the 2021–22 Teacher Follow-up Survey leaver rates (NCES, 2024); the replication rows apply the retrospective definition of Harris and Adams (2007) to the 2022–2024 March supplements, with teachers identified by the longest job of the previous calendar year and leaving defined by the survey-week occupation. [†] Total 2021–22 Teacher Follow-up Survey sample, current and former public and private school teachers together, drawn from the 2020–21 National Teacher and Principal Survey respondents (NCES, 2024). Harris and Adams and Aldeman and Yi do not report sample sizes. Source: author's calculations from linked CPS pairs and March supplements, 2005–2025.

The retrospective CPS estimates sit below all of this, and I can reproduce them too. Applying the definition of Harris and Adams (2007) to the 2022–2024 March supplements, a teacher by the longest job of the previous calendar year and a leaver by the survey-week occupation, yields 6.7 percent, against the 7.6 that Aldeman and Yi (2025) obtain pooling 2015–2024. The instrument is compressed on exactly one margin, occupation switching, which it records at 1.9 percentage points per year against the roughly 12 points that appear when two actual interviews are linked; exits from employment are similar under both instruments. Table A4 shows that the same compression appears in every profession, and that for the generic titles of accountants and social workers the linked levels are further inflated by independent re-coding of detailed occupations, which is why Figure 7 plots only the five professions with distinctive titles.

Table A4: Eight professions under the linked and the retrospective instruments.

	Linked design, 2005–2024			Retrospective, 2022–2024	
	Left detailed occupation	Left occupational field	Obser-vations	Left detailed occupation	Sample
Teachers	17.4	13.8	174,872	6.7	7,909
Registered nurses	20.2	17.4	75,160	7.5	5,176
Pharmacists	18.6	15.8	10,449	10.1	486
Physical therapists	21.8	15.9	8,385	11.5	440
Physicians	15.7	10.8	33,367	8.3	1,440
Lawyers	13.6	12.6	45,294	4.5	1,695
Social workers	41.3	33.3	22,874	15.9	1,170
Accountants and auditors	38.8	32.4	49,673	5.6	2,088

Notes: Annual exit rates in percent, weighted. The linked columns come from 12-month occupation pairs built, linked and validated exactly as the teacher panel, pooled over 2005–2024 baselines, under the conventional definition; field exit counts only moves outside the whole occupational field. The retrospective columns replicate the design of Harris and Adams (2007) and Aldeman and Yi (2025) on the 2022–2024 March supplements, comparing the occupation of the longest job held the previous calendar year with the survey-week occupation, without a degree restriction, weighted by supplement weights. Source: author's calculations from linked CPS pairs and March supplements.

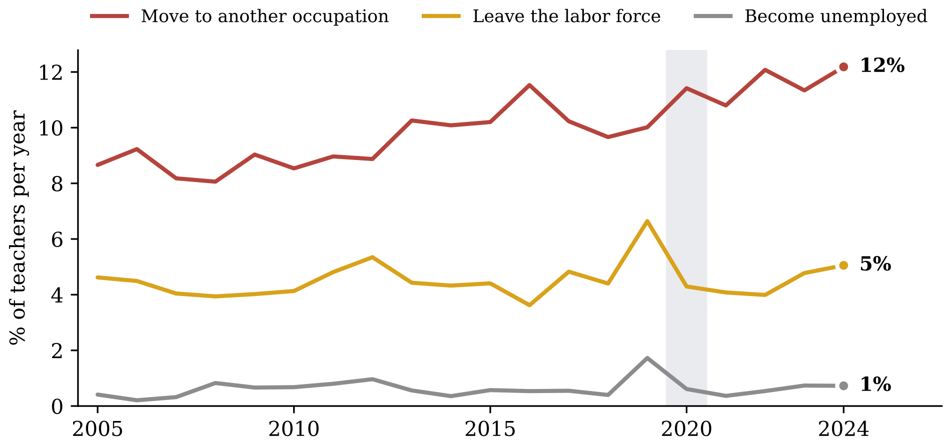
Appendix B. Estimation output

Table A5: Stayers vs non-returning leavers in the analytic sample (weighted means).

	All teachers	Stayers	Non-returning leavers	Difference
Age, years	44.0	43.5	46.7	3.15***
Female	76.8%	76.8%	77.0%	0.2
Female aged 25–44	37.8%	38.8%	31.7%	-7.1***
Married	72.0%	72.5%	69.5%	-3.0***
Number of own children (<18) at home	0.8	0.9	0.6	-0.22***
Child under 6 at home	18.1%	18.9%	13.9%	-5.0***
New baby during the year	2.5%	2.5%	2.8%	0.3
Black	8.1%	7.5%	11.6%	4.2***
Hispanic	7.9%	7.8%	8.5%	0.7*
Non-citizen	1.7%	1.5%	3.0%	1.5***
Master's degree or higher	52.9%	53.7%	48.4%	-5.3***
Professional degree or doctorate	2.8%	2.5%	4.4%	2.0***
Part-time (<35 h/week)	9.0%	7.2%	19.1%	11.9***
Holds more than one job	8.1%	7.9%	9.4%	1.5***
Public-sector employer	78.3%	80.5%	65.7%	-14.9***
Family income $75k+	76.4%	77.3%	71.4%	-5.9***
Elementary / middle school	61.8%	61.9%	61.2%	-0.7
Preschool / kindergarten	7.7%	7.1%	10.6%	3.5***
Secondary school	23.1%	23.5%	20.5%	-3.0***
Special education	7.5%	7.5%	7.6%	0.1
Observations	129,897	110,720	19,177	

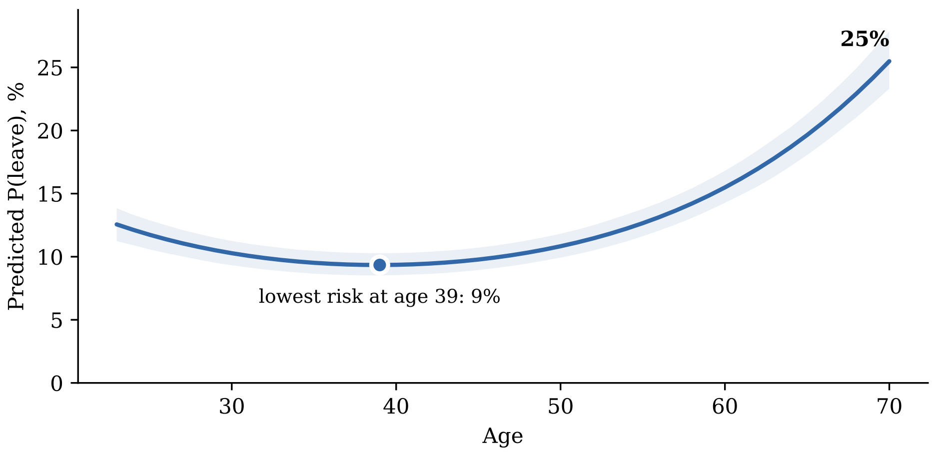
Notes: Weighted means at baseline over the linked analytic sample under the non-returning definition. Shaded rows mark differences that are statistically significant at the 5 percent level and at least two percentage points, or two years, wide. Standard errors are clustered by household. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A1: Leavers by destination over time.



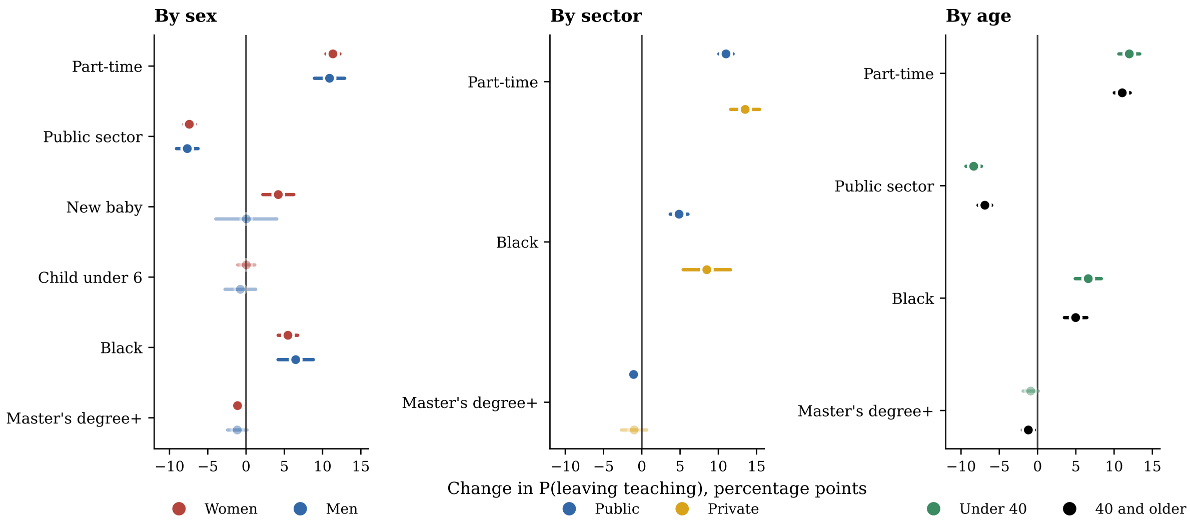
Notes: Non-returning leavers as a share of baseline teachers in each year, split by their situation twelve months later. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A2: Predicted probability of leaving by age.



Notes: Predicted probabilities from the probit of Table A8, with all covariates other than age and its square at their sample values. Source: author's calculations from linked CPS pairs, 2005–2025.

Figure A3: The same model estimated on subsamples.



Notes: Average marginal effects of the probit of Table A8 re-estimated on each subsample, with 95% confidence intervals and standard errors clustered by household. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A6: Exit routes. Probit average marginal effects by destination.

	(1) Any exit	(2) Another occupation	(3) Out of the labor force	(4) Unemployed
Part-time (<35 h/week)	11.31***	6.97***	3.47***	0.55***
	(0.42)	(0.35)	(0.22)	(0.07)
Public-sector employer	-7.49***	-6.40***	-0.65***	-0.27***
	(0.34)	(0.28)	(0.19)	(0.06)
Professional degree or doctorate	6.66***	6.34***	-0.18	0.17
	(0.84)	(0.66)	(0.48)	(0.15)
Master's degree or higher	-1.12***	-0.68**	-0.42**	-0.16***
	(0.31)	(0.26)	(0.17)	(0.06)
Black	5.68***	3.91***	1.25***	0.34***
	(0.57)	(0.47)	(0.31)	(0.09)
Non-citizen	5.72***	3.08***	1.88***	0.48***
	(1.11)	(0.90)	(0.56)	(0.15)
New baby during the year	-0.71	-0.58	-0.83	-0.13
	(2.01)	(1.62)	(1.51)	(0.40)
Female × new baby	5.11**	-2.33	6.58***	0.02
	(2.19)	(1.83)	(1.57)	(0.45)
Number of own children	-0.96***	-0.41***	-0.81***	-0.11***
	(0.18)	(0.15)	(0.12)	(0.04)
Female	-1.17***	-1.63***	0.72***	-0.21***
	(0.40)	(0.33)	(0.22)	(0.07)
Outgoing-rotation subsample with weekly earnings				
Log weekly earnings	-3.18***	-1.20***	-1.28***	-0.33***
	(0.33)	(0.27)	(0.18)	(0.08)
Base-year fixed effects	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes
Mean of dependent variable	0.148	0.097	0.045	0.006
Observations	129,897	129,897	129,897	129,897
wage subsample	44,823	44,823	44,823	44,823

Notes: Average marginal effects in percentage points from separate probits for any non-returning exit and for each destination, with the full covariate set and base-year fixed effects, standard errors clustered by household in parentheses. The wage row adds log weekly earnings on the outgoing-rotation subsample, where the conventional 12-month definition applies because those baselines have no re-interview window. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A7: Robustness to estimator and sample. Dependent variable is the non-returning leaver.

	(1) Probit (baseline)	(2) Logit	(3) Linear probability model	(4) Probit, excl. 2019–2021	(5) Probit, full-time only
Part-time (<35 h/week)	11.31***	10.78***	14.70***	10.75***	–
	(0.42)	(0.40)	(0.63)	(0.45)	
Public-sector employer	-7.49***	-7.33***	-8.43***	-7.50***	-7.27***
	(0.34)	(0.34)	(0.43)	(0.37)	(0.34)
Professional degree or doctorate	6.66***	6.48***	7.89***	6.38***	5.85***
	(0.84)	(0.80)	(1.12)	(0.90)	(0.84)
Master's degree or higher	-1.12***	-1.18***	-0.99***	-1.12***	-1.02***
	(0.31)	(0.32)	(0.31)	(0.33)	(0.31)
Black	5.68***	5.60***	6.42***	5.57***	5.34***
	(0.57)	(0.56)	(0.72)	(0.61)	(0.56)
Non-citizen	5.72***	5.53***	7.26***	5.76***	5.68***
	(1.11)	(1.05)	(1.54)	(1.18)	(1.14)
New baby during the year	-0.71	-0.91	-0.53	-0.51	-0.34
	(2.01)	(2.13)	(1.70)	(2.11)	(1.90)
Female $\times$ new baby	5.11**	5.31**	5.18***	5.15**	4.35**
	(2.19)	(2.28)	(1.97)	(2.29)	(2.08)
Number of own children	-0.96***	-1.04***	-0.65***	-0.87***	-0.84***
	(0.18)	(0.19)	(0.17)	(0.20)	(0.19)
Family income $75k+	-2.25***	-2.21***	-2.37***	-2.26***	-2.45***
	(0.36)	(0.36)	(0.39)	(0.38)	(0.36)
Female	-1.17***	-1.17***	-1.24***	-1.66***	-1.19***
	(0.40)	(0.40)	(0.42)	(0.42)	(0.39)
Base-year fixed effects	Yes	Yes	Yes	Yes	Yes
Full covariate set	Yes	Yes	Yes	Yes	Yes
Mean of dependent variable	0.148	0.148	0.148	0.145	0.131
Observations	129,897	129,897	129,897	113,751	118,039
Unique teachers	52,238	52,238	52,238	45,816	48,347

Notes: Average marginal effects in percentage points, standard errors clustered by household in parentheses. Column 4 drops the 2019–2021 baselines; column 5 keeps full-time teachers only, so the part-time and hours-not-reported indicators leave the model. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Source: author's calculations from linked CPS pairs, 2005–2025.

Table A8: Determinants of leaving teaching. Probit, dependent variable is the non-returning leaver.

	Leaves teaching with no observed return	
	(1) Probit coefficients	(2) Marginal effect (percentage points)
Age (years)	-0.053***	-1.14***
	(0.004)	(0.09)
Age^2/100	0.068***	1.46***
	(0.004)	(0.09)
Female	-0.055***	-1.17***
	(0.019)	(0.40)
Married	0.047***	1.01***
	(0.018)	(0.38)
Number of own children	-0.045***	-0.96***
	(0.009)	(0.18)
Child under 6 at home	-0.072*	-1.55*
	(0.041)	(0.87)
Female × child under 6	0.087**	1.87**
	(0.044)	(0.94)
New baby during the year	-0.033	-0.71
	(0.094)	(2.01)
Female × new baby	0.238**	5.11**
	(0.102)	(2.19)
Black	0.265***	5.68***
	(0.027)	(0.57)
Hispanic	0.067**	1.43**
	(0.029)	(0.62)
Non-citizen	0.267***	5.72***
	(0.052)	(1.11)
Master's degree or higher	-0.052***	-1.12***
	(0.015)	(0.31)
Professional degree or doctorate	0.311***	6.66***
	(0.039)	(0.84)
Part-time (<35 h/week)	0.528***	11.31***
	(0.020)	(0.42)
Hours not reported	0.352***	7.54***
	(0.022)	(0.47)
Holds more than one job	0.060***	1.28***
	(0.021)	(0.46)
Public-sector employer	-0.349***	-7.49***
	(0.016)	(0.34)
Family income $75k+	-0.105***	-2.25***
	(0.017)	(0.36)
Midwest	-0.082***	-1.76***
	(0.021)	(0.45)
South	0.007	0.15
	(0.020)	(0.42)
West	-0.017	-0.36
	(0.021)	(0.45)
Preschool / kindergarten	0.075***	1.60***
	(0.026)	(0.55)
Secondary school	-0.047***	-1.01***
	(0.018)	(0.38)
Special education	0.088***	1.89***
	(0.026)	(0.55)
Base-year fixed effects	Yes	Yes
Mean of dependent variable	0.148	0.148
Observations	129,897	129,897
Unique teachers	52,238	52,238
Pseudo R^2	0.068	

Notes: Column 1 reports probit coefficients and column 2 average marginal effects in percentage points, with standard errors clustered by household in parentheses. Base-year fixed effects included. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Source: author's calculations from linked CPS pairs, 2005–2025.